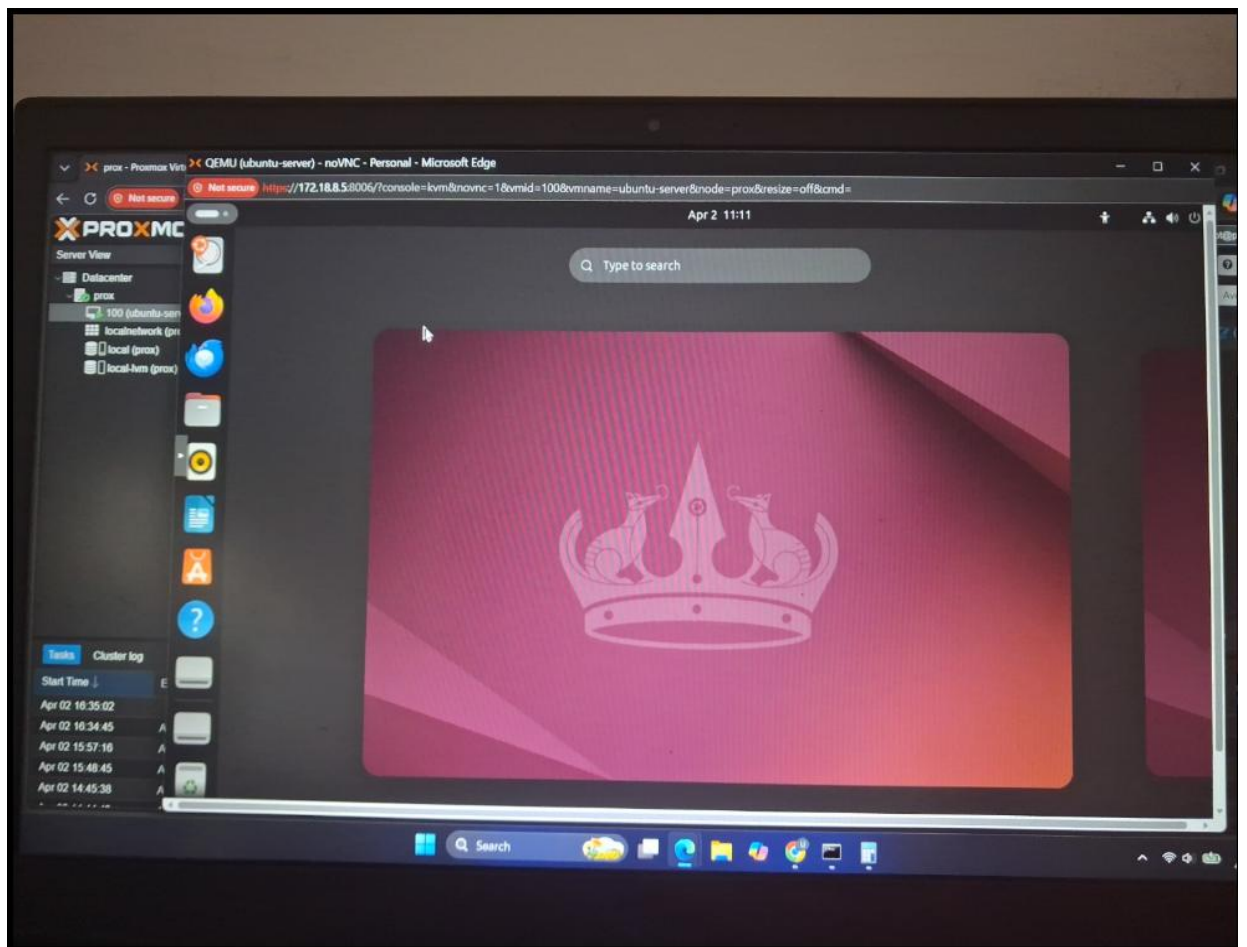


Install RADIUS and Configure on Ubuntu

Prerequisites

`sudo apt update`

`sudo apt upgrade`



Install Apache Web Server

`sudo apt -y install apache2`

`sudo systemctl enable --now apache2`

```
sudo ufw allow WWW
```

Install PHP and additional PHP Modules

```
sudo apt -y install php libapache2-mod-php php-{gd,common,mail,mail-  
mime,mysql,pear,db,mbstring,xml,curl}
```

Check the PHP version:

```
php -v
```

Create a file in /var/www/html called phpinfo.php (it can be any name, it doesn't matter)

```
sudo nano /var/www/html/phpinfo.php
```

Install MySQL

```
sudo apt -y install mysql-server
```

```
sudo mysql_secure_installation
```

Enter current password for root (enter for none):

Install FreeRADIUS and Configure with MySQL/MariaDB on Ubuntu

```
sudo apt -y install freeradius freeradius-mysql freeradius-utils -y
```

Test the FreeRADIUS Server

```
sudo systemctl stop freeradius
```

```
sudo freeradius -X
```

start and enable FreeRADIUS service so it runs on system boot:

```
sudo systemctl enable --now freeradius
```

Allow FreeRADIUS in Firewall

```
sudo ufw allow to any port 1812 proto udp
```

```
sudo ufw allow to any port 1813 proto udp
```

Configure FreeRADIUS to use

```
sudo mysql -u root -p
```

```
sudo su -
```

```
mysql -u root -p radius < /etc/freeradius/3.0/mods-config/sql/main/mysql/schema.sql
```

```
sudo su - edxd
```

```
sudo mysql -u root -p -e "use radius;show tables;"
```

```
sudo ln -s /etc/freeradius/3.0/mods-available/sql /etc/freeradius/3.0/mods-enabled/
```

```
sudo apt install nano
```

```
sudo nano /etc/freeradius/3.0/mods-enabled/sql
```

There's quite a bit of text, but most of it is commented out. We'll just need to edit a few things.

Change dialect = "sqlite" to dialect = "mysql"

Change driver = "rlm_sql_null" to driver = "rlm_sql_\${dialect}"

If we use MySQL the FreeRADIUS configuration assumes the use of TLS certs by default. For the purpose of this tutorial, we won't be using TLS certs, so we'll comment out the MySQL TLS section by adding a # sign in at the beginning of every line in the tls section. The TLS section looks something like this:

```
mysql {  
    # If any of the files below are set, TLS encryption is enabled  
  
    tls {  
        ca_file = "/etc/ssl/certs/my_ca.crt"  
        ca_path = "/etc/ssl/certs/"
```

```

certificate_file = "/etc/ssl/certs/private/client.crt"
private_key_file = "/etc/ssl/certs/private/client.key"
cipher = "DHE-RSA-AES256-SHA:AES128-SHA"
tls_required = yes
tls_check_cert = no
tls_check_cert_cn = no
}

# If yes, (or auto and libmysqlclient reports warnings are
# available), will retrieve and log additional warnings from
# the server if an error has occurred. Defaults to 'auto'
warnings = auto
}

```

And this is how it looks with the TLS section commented out:

```

mysql {
    # If any of the files below are set, TLS encryption is enabled
    # tls {
    #     ca_file = "/etc/ssl/certs/my_ca.crt"
    #     ca_path = "/etc/ssl/certs/"
    #     certificate_file = "/etc/ssl/certs/private/client.crt"
    #     private_key_file = "/etc/ssl/certs/private/client.key"
    #     cipher = "DHE-RSA-AES256-SHA:AES128-SHA"
    #     tls_required = yes
    #     tls_check_cert = no
    #     tls_check_cert_cn = no
    # }
}

```

```
#}  
  
# If yes, (or auto and libmysqlclient reports warnings are  
# available), will retrieve and log additional warnings from  
# the server if an error has occurred. Defaults to 'auto'  
  
warnings = auto  
  
}
```

Next, we'll uncomment the Connection info section and add in the connection details to our MySQL/MariaDB database. First, uncomment (remove the # signs) from the beginning of the lines starting with server, port, login, and password.

server – this is the server where the database is located. In this case, it's the local server, so we can leave "localhost"

port – is set to 3306, which is the default port for the classic MySQL protocol. Leave it as is unless you have changed the MySQL port.

login – this is the database user you created earlier for FreeRADIUS to use. I created the user radius, so I'll leave it as is. You change it if your user is something else.

password – the password for that MySQL user that you also set earlier. This is its initial state:

```
# Connection info:
```

```
#
```

```
# server = "localhost"
```

```
# port = 3306
```

```
# login = "radius"
```

```
# password = "radpass"
```

And here it is edited.

```
# Connection info:
```

```
#
```

```
server = "localhost"
```

```
port = 3306
```

```
login = "radius"
```

```
password = "Somestrongpassword_321"
```

A few lines lower, we need to configure the name of the database. By default, it looks like this:

```
# Database table configuration for everything except Oracle
```

```
radius_db = "radius"
```

Instead of radius, input the database you created. Since I created the database radius, I'll leave it as is:

```
# Database table configuration for everything except Oracle
```

```
radius_db = "radius"
```

Further down, we'll uncomment a line containing `read_clients = yes`. This is to enable FreeRADIUS to read clients from the database. Here is how it looks:

```
# Set to 'yes' to read radius clients from the database ('nas' table)
```

```
# Clients will ONLY be read on server startup.
```

```
# read_clients = yes
```

Just remove the `#` sign to uncomment it:

```
# Set to 'yes' to read radius clients from the database ('nas' table)
```

```
# Clients will ONLY be read on server startup.
```

```
read_clients = yes
```

Just a few lines lower, we want `client_table = "nas"` to be uncommented. It should be uncommented by default, but just check to make sure it looks like this:

```
# Table to keep radius client info
```

```
client_table = "nas"
```

Now change the group rights of the file we just edited:

```
sudo chgrp -h freerad /etc/freeradius/3.0/mods-available/sql
```

```
sudo chown -R freerad:freerad /etc/freeradius/3.0/mods-enabled/sql
```

And restart the FreeRADIUS service:

```
sudo systemctl restart freeradius.service
```

Since we've done quite a few edits, we should run FreeRADIUS in debug mode so we know if we made any mistake, before going further.

First, stop the FreeRADIUS service since we can't have 2 instances of the service running simultaneously:

```
sudo systemctl stop freeradius.service
```

And run FreeRADIUS in debug mode:

```
sudo freeradius -X
```

The output looks something like this:

```
Listening on auth address * port 1812 bound to server default
```

```
Listening on acct address * port 1813 bound to server default
```

```
Listening on auth address :: port 1812 bound to server default
```

```
Listening on acct address :: port 1813 bound to server default
```

```
Listening on auth address 127.0.0.1 port 18120 bound to server inner-tunnel
```

```
Listening on proxy address * port 52025
```

```
Listening on proxy address :: port 42807
```

Ready to process requests

Exit debug mode by pressing Ctrl+C and then start FreeRADIUS again by running:

```
sudo systemctl start freeradius.service
```

Install & Configure daloRADIUS (FreeRADIUS GUI) on Ubuntu

```
sudo apt -y install wget unzip
```

Now download daloRADIUS and cd into the newly created daloradius-1.3 folder:

```
wget https://github.com/lirantal/daloradius/archive/1.3.zip
```

```
unzip 1.3.zip
```

```
cd daloradius-1.3
```

Populate the database with the daloRADIUS schema:

```
sudo mysql -u root -p radius < contrib/db/fr2-mysql-daloradius-and-freeradius.sql
```

```
sudo mysql -u root -p radius < contrib/db/mysql-daloradius.sql
```

cd out of the daloradius-1.3 directory, and move the folder into the document root as daloradius:

```
cd ..
```

```
sudo mv daloradius-1.3 /var/www/html/daloradius
```

Next, we'll change the owner and group for the daloradius folder to www-data:www-data, which are the user and group under which the Apache Web Server runs.

```
sudo chown -R www-data:www-data /var/www/html/daloradius/
```

Now, we'll need to create the daloRADIUS configuration file. Right now, we're just provided a sample file, so we'll make a copy from that sample file:

```
sudo cp /var/www/html/daloradius/library/daloradius.conf.php.sample  
/var/www/html/daloradius/library/daloradius.conf.php
```

We'll also change the permissions for the daloRADIUS configuration file:

```
sudo chmod 664 /var/www/html/daloradius/library/daloradius.conf.php
```

Next, we edit a few variables in the daloRADIUS connection file so it can connect to the FreeRADIUS database.

Open the configuration file using your favorite editor:

```
sudo nano /var/www/html/daloradius/library/daloradius.conf.php
```

Similarly to what we've done earlier when editing the FreeRADIUS config file, we need to adjust the variables for the database user, their password, and the database name. Those are all the edits for the scope of this tutorial.

This is how they initially look in the daloRADIUS configuration file:

```
$configValues['CONFIG_DB_USER'] = 'root';  
$configValues['CONFIG_DB_PASS'] = '';  
$configValues['CONFIG_DB_NAME'] = 'radius';
```

This is how it looks after editing the details for the database I created earlier:

```
$configValues['CONFIG_DB_USER'] = 'radius';
```

```
$configValues['CONFIG_DB_PASS'] = 'Sometrangepassword_321';
```

```
$configValues['CONFIG_DB_NAME'] = 'radius'
```

Lastly, restart FreeRADIUS and Apache to make sure everything works:

```
sudo systemctl restart freeradius.service apache2
```

Access daloRADIUS

http://server_ip_address/daloradius

Example:

`http://172.18.8.16:8080//daloradius`

